

Percentage Decrease

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A **percentage decrease** describes how much a quantity is reduced compared to its original value.

Amount of Decrease

To find the amount of decrease, multiply the original value by the percent decrease.

$$\text{Decrease} = \text{Original Value} \times \frac{\% \text{ Decrease}}{100}$$

New Value

To find the new value after a decrease, subtract the amount of decrease from the original value.

$$\text{New Value} = \text{Original Value} - \text{Decrease}$$

Examples

Example 1

A jacket originally costs ₱1,200. It is marked down by 15%. What is the sale price?

$$\text{Decrease} = 1200(0.15) = \text{₱}180$$

$$\text{Sale Price} = 1200 - 180 = \text{₱}1,020$$

Example 2

The enrollment of a school decreased from 800 students to 680 students. What is the percentage decrease?

$$\text{Decrease} = 800 - 680 = 120$$

$$\text{Percent Decrease} = \frac{120}{800} \times 100 = 15\%$$

Example 3

A bicycle worth ₱8,500 is sold at a 20% discount. How much is the discount?

$$\text{Discount} = 8500(0.20) = \text{₱}1,700$$

Practice Exercises

Practice 1

A town's population decreased from 12,000 to 10,800. Find the percentage decrease.

Practice 2

A television priced at ₱25,000 is discounted by 12%. What is its new price?

Practice 3

A farmer harvested 3,500 kg of rice last year and 2,975 kg this year. What percent decrease occurred?

Practice 4

A computer costing ₱32,000 is offered with a 25% discount. What is the final price?

Practice 5

The number of books in a library decreased from 4,500 to 4,050. Find the percentage decrease.

Practice 6

A pair of shoes costs ₱2,400. During a sale, the price is reduced by 18%. How much will the customer pay?

Practice 7

The water level in a reservoir dropped from 150 meters to 120 meters. What is the percentage decrease?

Activities

Activity 1

A jacket originally costs ₱1,500. It is marked down by 10%. What is the sale price?

Activity 2

The enrollment of a school decreased from 900 students to 765 students. What is the percentage decrease?

Activity 3

A bicycle worth ₱9,200 is sold at a 15% discount. How much is the discount?

Activity 4

A town's population decreased from 15,000 to 13,500. Find the percentage decrease.

Activity 5

A television priced at ₱28,000 is discounted by 8%. What is its new price?

Activity 6

A farmer harvested 4,000 kg of rice last year and 3,400 kg this year. What percent decrease occurred?

Activity 7

A computer costing ₱36,000 is offered with a 30% discount. What is the final price?

Activity 8

The number of books in a library decreased from 5,200 to 4,420. Find the percentage decrease.

Activity 9

A pair of shoes costs ₱3,000. During a sale, the price is reduced by 22%. How much will the customer pay?

Activity 10

The water level in a reservoir dropped from 180 meters to 144 meters. What is the percentage decrease?

Thank you!